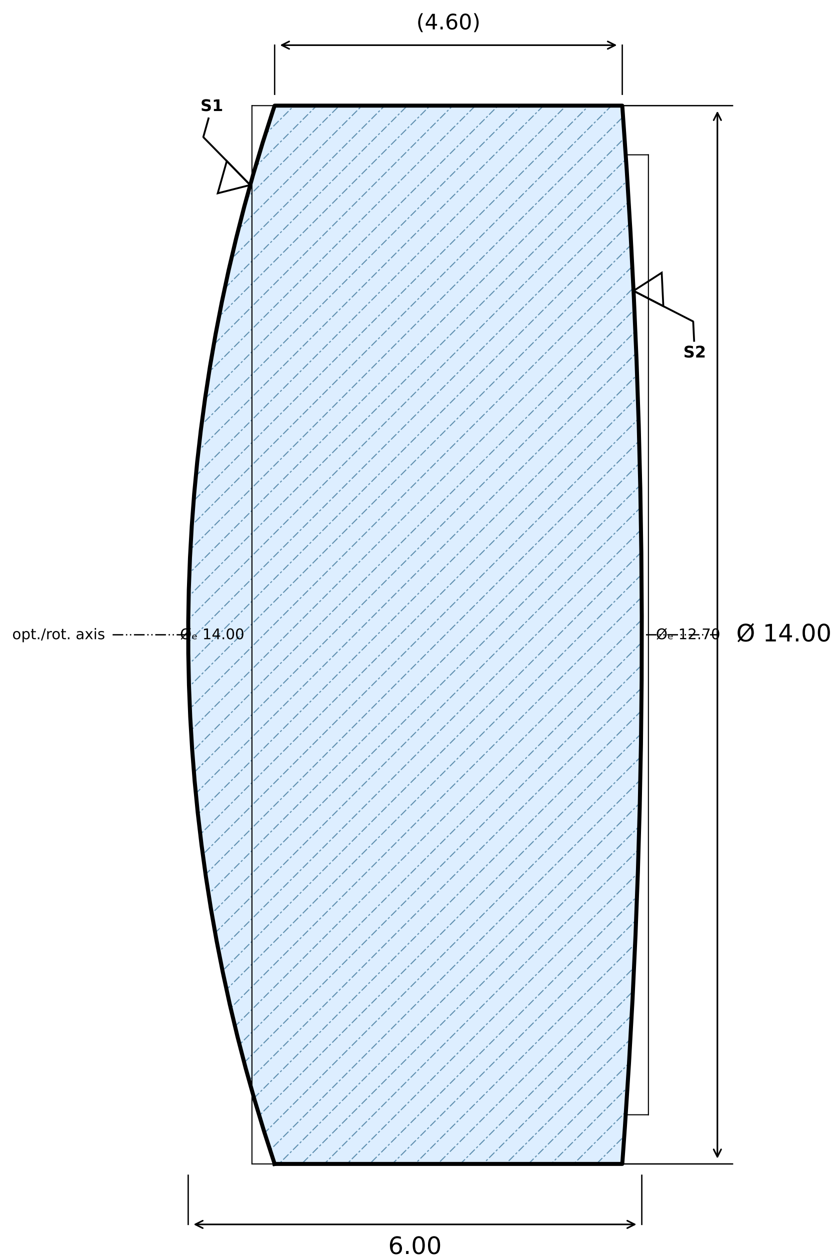
f′ = 18.85 mm



Ang. nach ISO 10110; λ = 546.07 nm

SURFACE 1 (FRONT)	MATERIAL	SURFACE 2 (REAR)
R +22.000 CX	N-BK7 (SCHOTT)	R -95.000 CX
$\varnothing_e$ 14.00	n_d = 1.5168 (587.6 nm)	$\varnothing_e$ 12.70
3/ (0.5/0.25); λ=546.1 nm	ν_d = 64.17 (587.6 nm)	3/ (1); λ=546.1 nm
4/ 0.5′	0/ 10 nm/cm	4/ 0.5′
5/ 2×0.04	1/ 1×0.16	5/ 2×0.063
6/ —	2/ NH010; A	6/ —
λ 7/ AR 400-700 nm, R<0.5%		λ 7/ AR 400-700 nm, R<0.5%
13/ —		13/ —

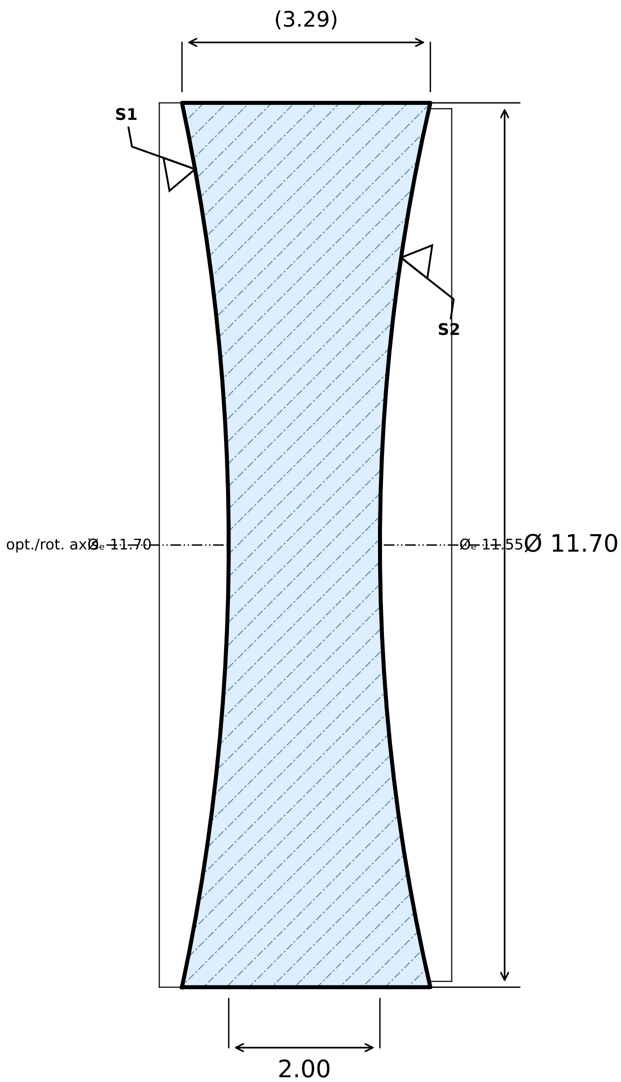
5-Element Photographic Objective OPT-001	B. Lutzer	K. Harrison	2026-07-16	10:1	1 / 5	A
ORGANISATION / PROJECT / PART NO.	DRAWN BY	APPROVED	DATE	SCALE	SHEET	REV

NOTES

DIM. IN mm

GENERAL TOL. PER ISO 10110-11
Ø±0.10 CT±0.15 3/ 5 4/ 3′ 5/ 2×0.063

f′ = 18.85 mm



Ang. nach ISO 10110; λ = 546.07 nm

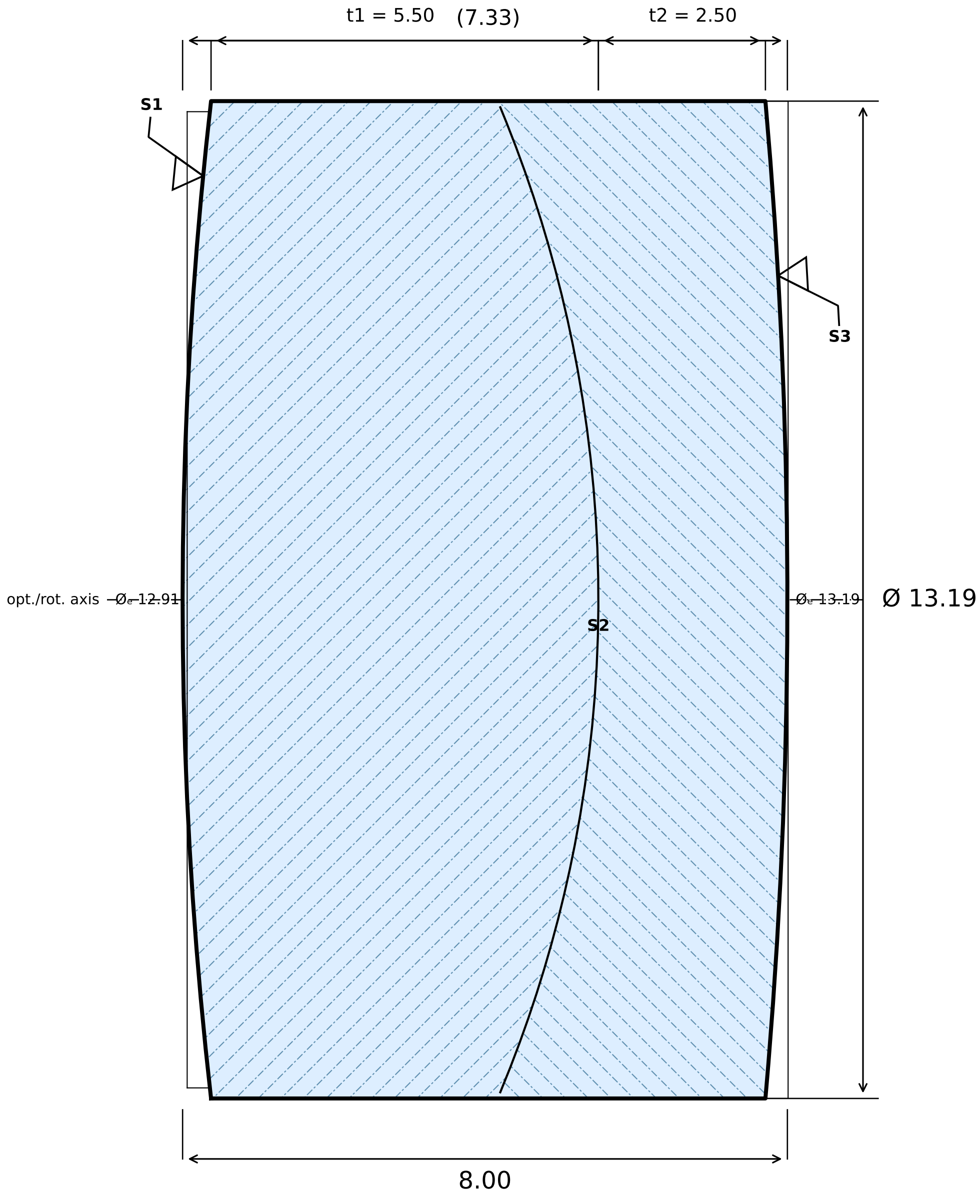
SURFACE 1 (FRONT)	MATERIAL	SURFACE 2 (REAR)
R -28.000 CC	N-SF2	R +26.000 CC
Øₑ 11.70	n_d = 1.6477 (587.6 nm)	Øₑ 11.55
3/ (0.5); λ=546.1 nm	ν_d = 33.82 (587.6 nm)	3/ (0.5); λ=546.1 nm
4/ 1′	0/ 10 nm/cm	4/ 1′
5/ 2×0.04	1/ 1×0.16	5/ 2×0.04
6/ —	2/ NH010; B	6/ —
λ 7/ AR 400-700 nm, R<0.5%		λ 7/ AR 400-700 nm, R<0.5%
13/ —		13/ —

5-Element Photographic Objective OPT-002	B. Lutzer	K. Harrison	2026-07-16	10:1	2 / 5	A
ORGANISATION / PROJECT / PART NO.	DRAWN BY	APPROVED	DATE	SCALE	SHEET	REV

NOTES

DIM. IN mm
GENERAL TOL. PER ISO 10110-11
Ø±0.10 CT±0.15 3/ 5 4/ 3′ 5/ 2×0.063

f′ = 18.85 mm



Ang. nach ISO 10110; λ = 546.07 nm

SURFACE 1 (FRONT)	GLASS 1	SURFACE 2 (CEMENT)	GLASS 2	SURFACE 3 (REAR)
R +58.000 CX Øₑ 12.91 3/ (0.5); λ=546.1 nm 4/ 0.5′ 5/ 2×0.04 6/ — λ 7/ AR 400-700 nm, R<0.3% 13/ —	N-BAK1 (SCHOTT) n_d = 1.5725 (587.6 nm) ν_d = 57.55 (587.6 nm) 0/ 6 nm/cm 1/ 1×0.10 2/ NH010; A	R -17.000 CC Øₑ 13.03 3/ (0.25); λ=546.1 nm 4/ 0.5′ 5/ 5×0.016 6/ — λ 7/ — 13/ — 15/ —	N-SF5 (SCHOTT) n_d = 1.6727 (587.6 nm) ν_d = 32.25 (587.6 nm) 0/ 6 nm/cm 1/ 1×0.10 2/ NH010; A	R -75.000 CX Øₑ 13.19 3/ (0.5); λ=546.1 nm 4/ 0.5′ 5/ 2×0.04 6/ — λ 7/ AR 400-700 nm, R<0.3% 13/ —

5-Element Photographic Objective OPT-003	**B. Lutzer**	**K. Harrison**	**2026-07-16**	**10:1**	**3 / 5**	**A**
ORGANISATION / PROJECT / PART NO.	DRAWN BY	APPROVED	DATE	SCALE	SHEET	REV

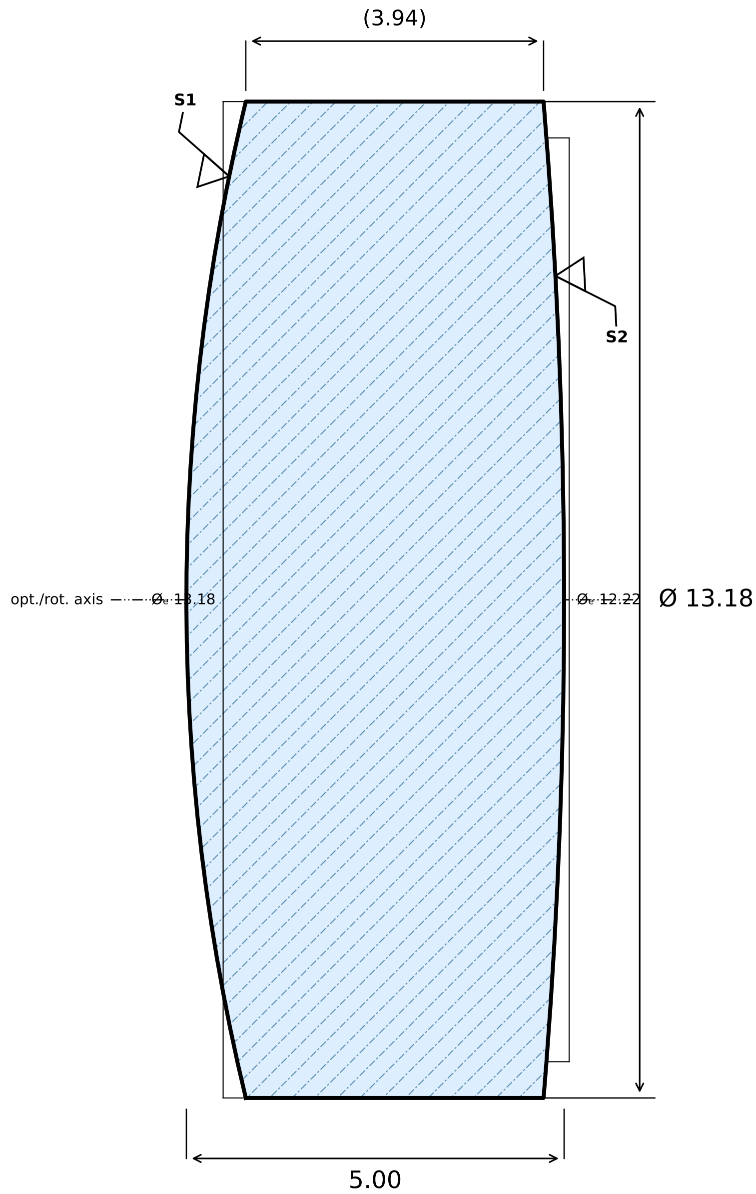
NOTES

Cement with NOA-61 or equivalent UV adhesive.

DIM. IN mm

GENERAL TOL. PER ISO 10110-11
Ø±0.10 CT±0.15 3/ 5 4/ 3′ 5/ 2×0.063

f′ = 18.85 mm



Ang. nach ISO 10110; λ = 546.07 nm

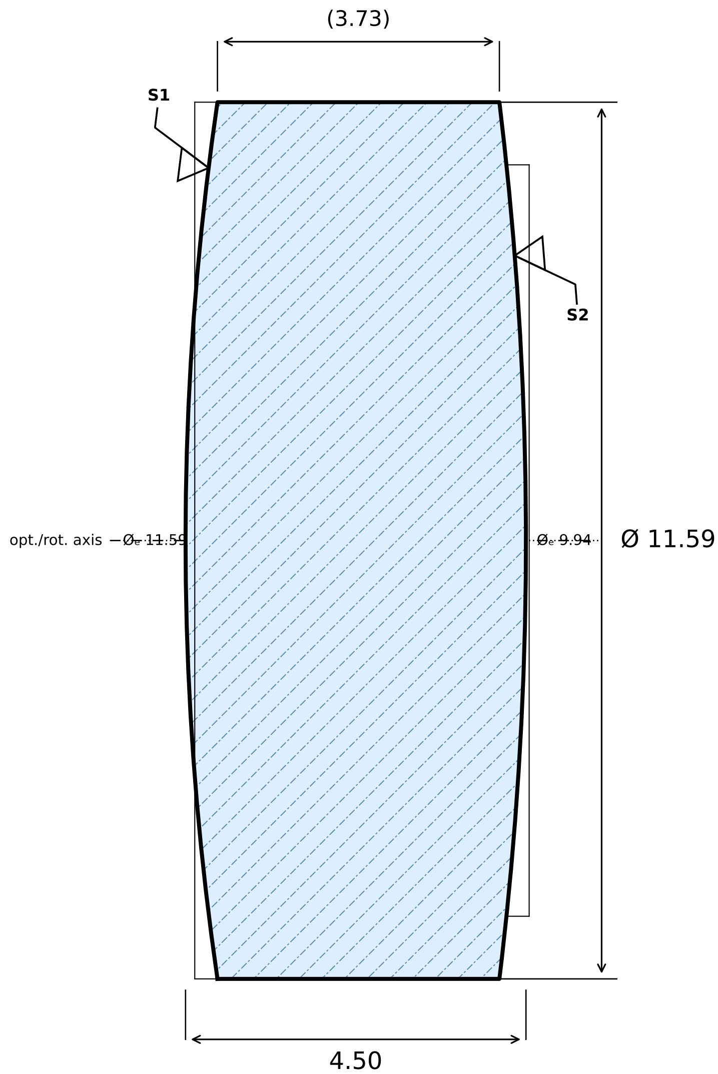
SURFACE 1 (FRONT)	MATERIAL	SURFACE 2 (REAR)
R +28.000 CX	N-SSK5	R -80.000 CX
Øₑ 13.18	n_d = 1.6584 (587.6 nm)	Øₑ 12.22
3/ (0.5); λ=546.1 nm	ν_d = 50.88 (587.6 nm)	3/ (1); λ=546.1 nm
4/ 0.5′	0/ 10 nm/cm	4/ 0.5′
5/ 2×0.063	1/ 1×0.16	5/ 2×0.063
6/ —	2/ NH004; B	6/ —
λ 7/ AR 400-700 nm, R<0.5%		λ 7/ AR 400-700 nm, R<0.5%
13/ —		13/ —

5-Element Photographic Objective OPT-004	B. Lutzer	—	2026-07-16	10:1	4 / 5	A
ORGANISATION / PROJECT / PART NO.	DRAWN BY	APPROVED	DATE	SCALE	SHEET	REV

NOTES

DIM. IN mm
GENERAL TOL. PER ISO 10110-11
Ø±0.10 CT±0.15 3/ 5 4/ 3′ 5/ 2×0.063

f′ = 18.85 mm



Ang. nach ISO 10110; λ = 546.07 nm

SURFACE 1 (FRONT)	MATERIAL	SURFACE 2 (REAR)
R +40.000 CX	N-LAK22	R -48.000 CX
Øₑ 11.59	n_d = 1.6511 (587.6 nm)	Øₑ 9.94
3/ (0.5); λ=546.1 nm	ν_d = 55.89 (587.6 nm)	3/ (0.5); λ=546.1 nm
4/ 0.5′	0/ 10 nm/cm	4/ 0.5′
5/ 2×0.04	1/ 1×0.16	5/ 2×0.04
6/ —	2/ NH010; A	6/ —
λ 7/ AR 400-700 nm, R<0.3%		λ 7/ AR 400-700 nm, R<0.3%
13/ —		13/ —

5-Element Photographic Objective OPT-005	B. Lutzer	—	2026-07-16	10:1	5 / 5	A
ORGANISATION / PROJECT / PART NO.	DRAWN BY	APPROVED	DATE	SCALE	SHEET	REV

NOTES

DIM. IN mm
GENERAL TOL. PER ISO 10110-11
Ø±0.10 CT±0.15 3/ 5 4/ 3′ 5/ 2×0.063